

1. (Fall 2024 #3) Find the derivative of the following functions. You do not need to simplify your answer.

(a) $f(x) = \tan\left(\frac{e^{\sqrt{x}}}{\ln x}\right)$

(b) $f(x) = (\sin x)^{\cos^2 x}$

2. (Spring 2024, #3) Find the derivatives of the functions given below. You do not need to simplify the final form of your derivative.

(a)

$$g(x) = \frac{x}{(x^2 + \tan(x))^5}$$

(b)

$$f(x) = x^2 \cdot \cos(\ln(x^3 - 1) + 2)$$

3. (Spring 2023, #1) Find the derivatives of the following functions. You do not need to simplify your answers.

(a) $f(x) = x^{e^x}$

(b) $h(x) = \frac{\cos(x^3 + 2x)\sqrt{x}}{\tan(x)}$

4. (Fall 2022, #2)

(a) Find every value of k such that the following function is continuous everywhere:

$$f(x) = \begin{cases} kx - 4 & x \leq 2 \\ x^3 & x > 2 \end{cases}$$

(b) Let

$$g(x) = e^{\cos(\log_7(x))}$$

Use derivative rules to find $g'(x)$.

(c) Let

$$g(x) = \frac{1}{x+1}$$

Find $g'(1)$ using the limit definition of the derivative.

5. (Spring 2022, #2) Compute the derivatives of the following functions. Do not simplify your answer.

(a) $f(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

(b) $y = \tan(x)^{2x}$